EE115B *Digital Circuits*

Instructor: Chenxi Xiao

Part of slides are from

© Pearson Education

Hengzhao Yang

Zhifeng Zhu

Yajun Ha

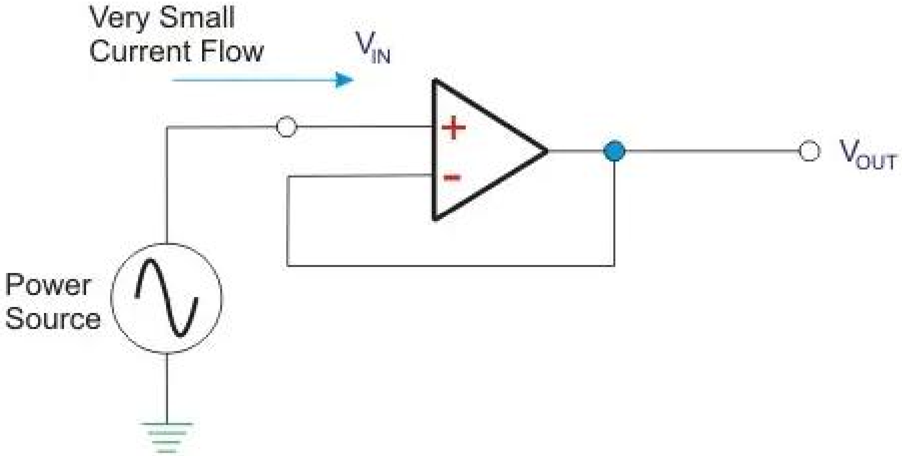
上海科技大学

ShanghaiTech University

ADC/DAC

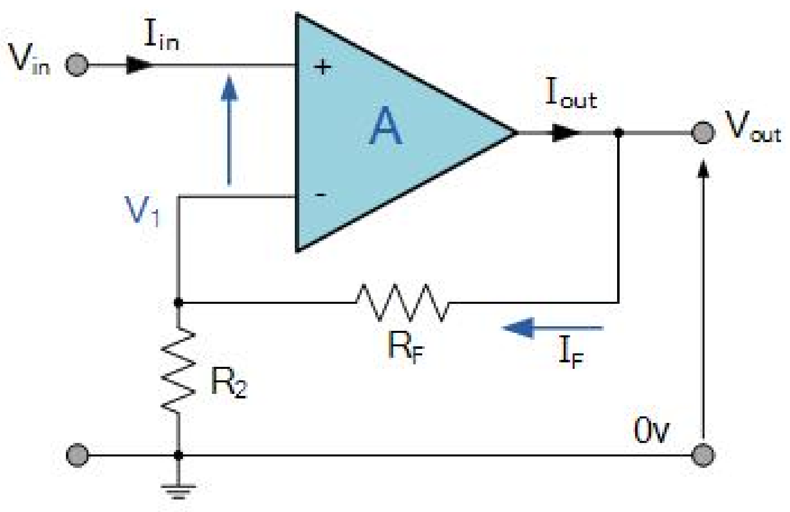
Voltage follower

- Vout=Vin
- Makes it possible to design circuits before/after amplifiers separately.
- Commonly used as output buffer (reduce output impedance)



Non-inverting amplifier

In non-inverting operational amplifier configuration, the input voltage signal, (V_{in}) is applied directly to the non-inverting (+) input terminal .



$$V_1 = \frac{R_2}{R_2 + R_F} \times V_{OUT}$$

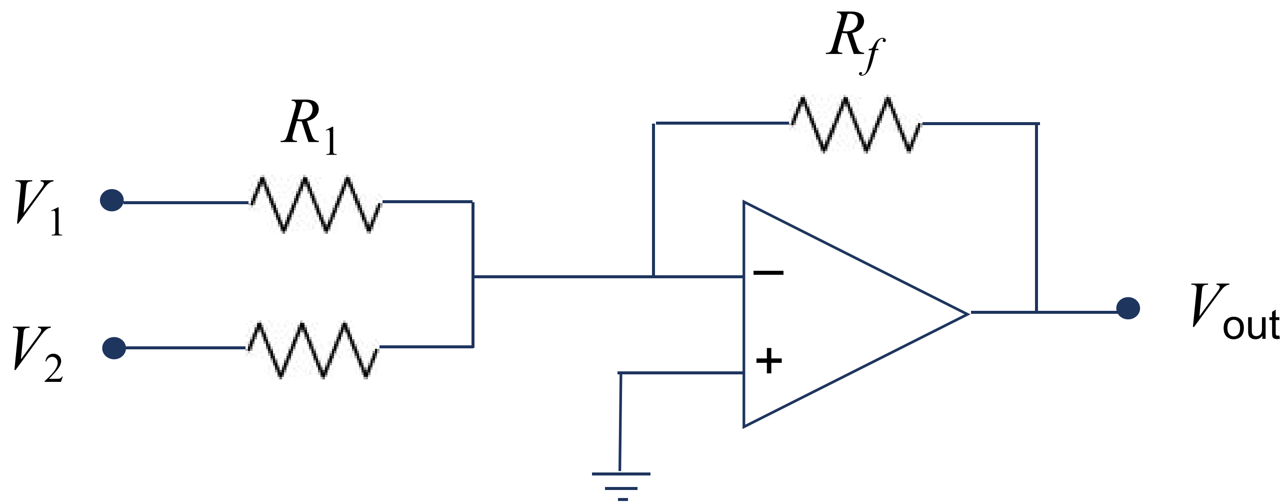
Ideal Summing Point: $V_1 = V_{IN}$

Voltage Gain, $A_{(V)}$ is equal to: $\frac{V_{OUT}}{V_{IN}}$

Then, $A_{(V)} = \frac{V_{OUT}}{V_{IN}} = \frac{R_2 + R_F}{R_2}$

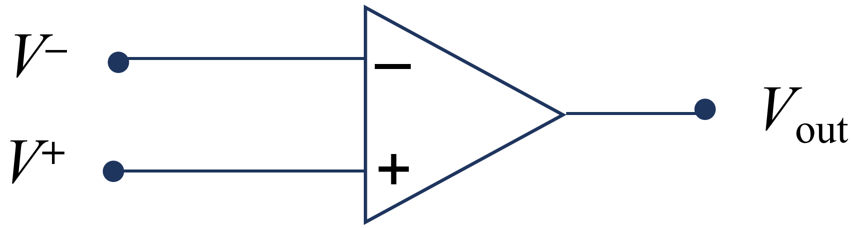
Transpose to give: $A_{(V)} = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_F}{R_2}$

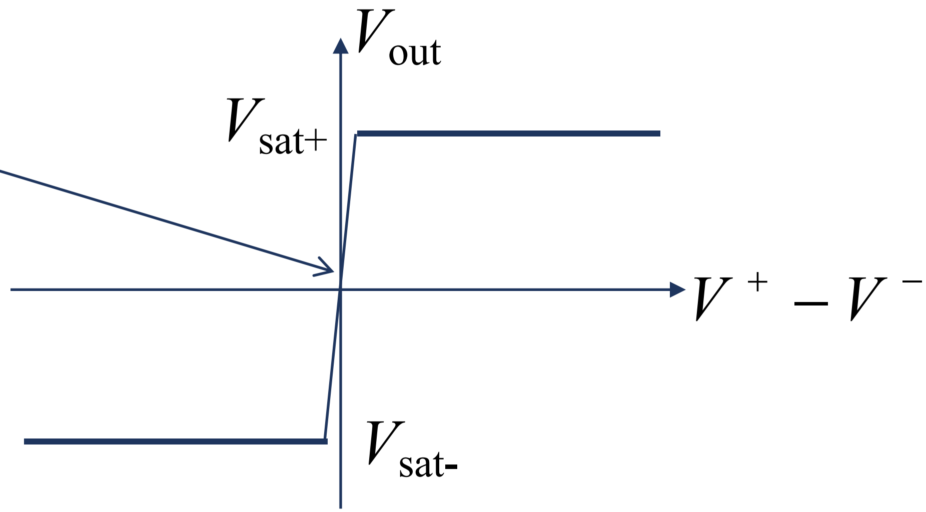
Inverting amplifier (with sum)



$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2\right)$$

Review of Op-Amps





<u>In the linear region</u>

$$V_{out} = A(V^+ - V^-)$$

A is huge (10^5 or larger) for DC component

<u>In saturation</u>

$$V_{out} = \begin{cases} V_{sat+} & \text{if } V^+ > V^- \\ V_{sat-} & \text{if } V^+ < V^- \end{cases}$$

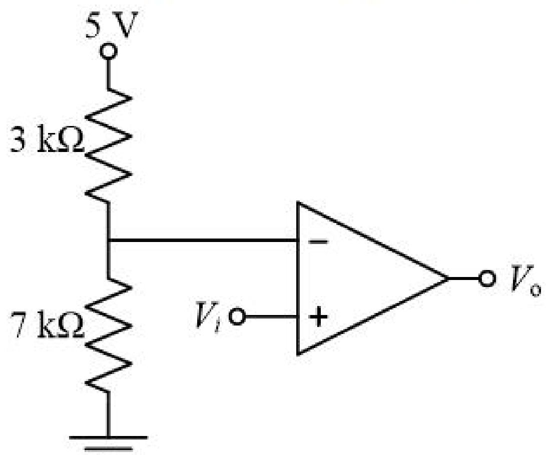
(V_{sat+} and V_{sat-} are largely determined by power supplies.)

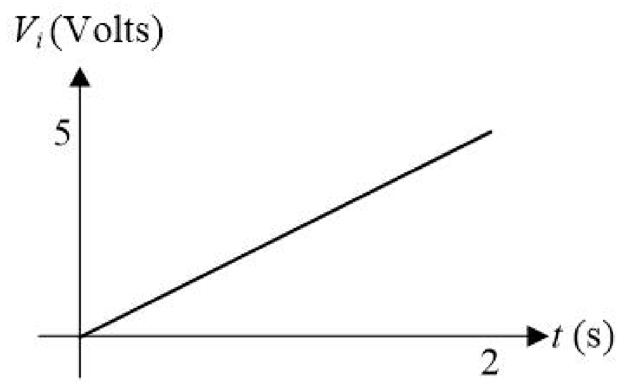
$\rightarrow$ essentially **digital output** because the linear region is tiny

Review of Op-Amps

Using an Op Amp as a Comparator

In the circuit below, the analog voltage V_i varies linearly from 0 V to 5 V. Graph the output, V_o, from $t = 0$ to $t = 2$ s. Assume the positive and negative saturation voltages of the op amp are +15 V and -15 V, respectively.





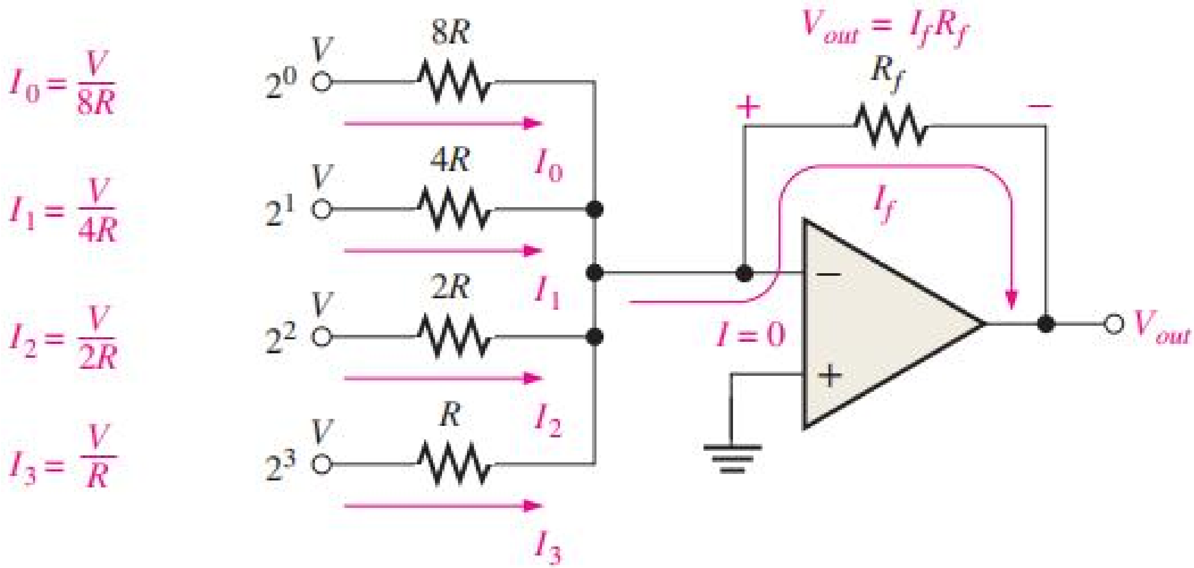
Solution

$$V^- = \text{voltage at inverting terminal} = \left(\frac{7\,\text{k}\Omega}{7\,\text{k}\Omega + 3\,\text{k}\Omega}\right)5 = 3.5\text{ V}$$

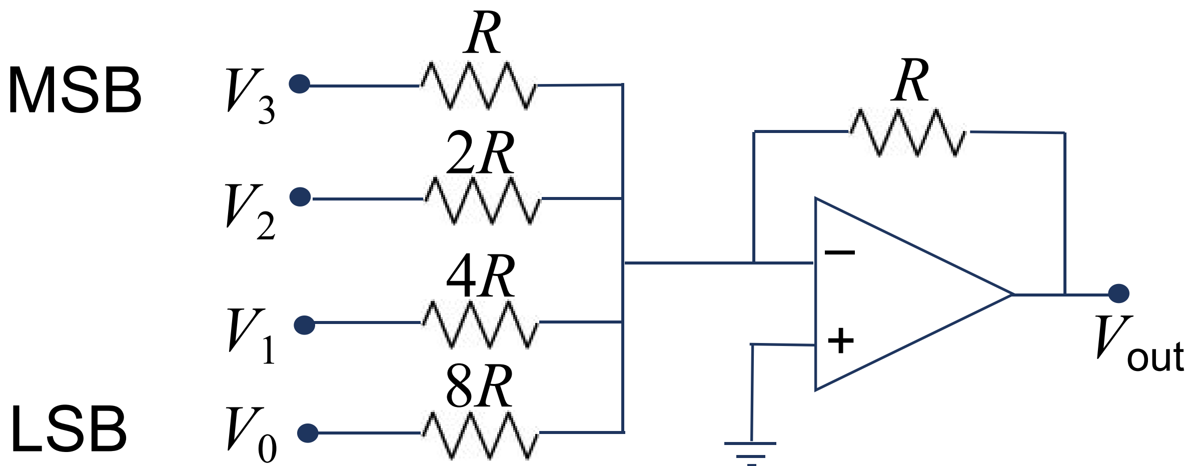
$$\Rightarrow \quad V_o \approx \begin{cases} -15\text{V}, & \text{if } V_i < 3.5\text{V} \\ +15\text{V}, & \text{if } V_i > 3.5\text{V} \end{cases} \quad \Rightarrow \quad V_o \approx \begin{cases} -15\text{V}, & 0 < t < 1.4\text{s} \\ +15\text{V}, & 1.4 < t < 2\text{s} \end{cases}$$

DAC: Binary-Weighted-Input DAC

Uses a resistor network with resistance values that represent the binary weights of the input bits of the digital code.



Example: 4-bit ADC



Binary input: $b_3b_2b_1b_0$

Assume

$$V_i = \begin{cases} V_H, & \text{if } b_i=1 \\ 0, & \text{if } b_i=0 \end{cases}$$

Equivalently, $V_i = b_i V_H$

Goal: Verify that this is a DAC (verify that V_{out} is proportional to input word value).

$$V_{out} = -\left(V_3 + \frac{V_2}{2} + \frac{V_1}{4} + \frac{V_0}{8} \right) =$$

$$= -V_H/8(2^3b_3+2^2b_2+2^1b_1+2^0b_0)$$

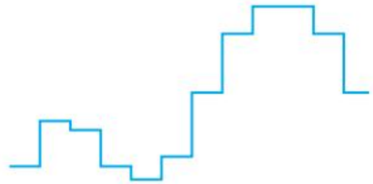
VH is high logic voltage

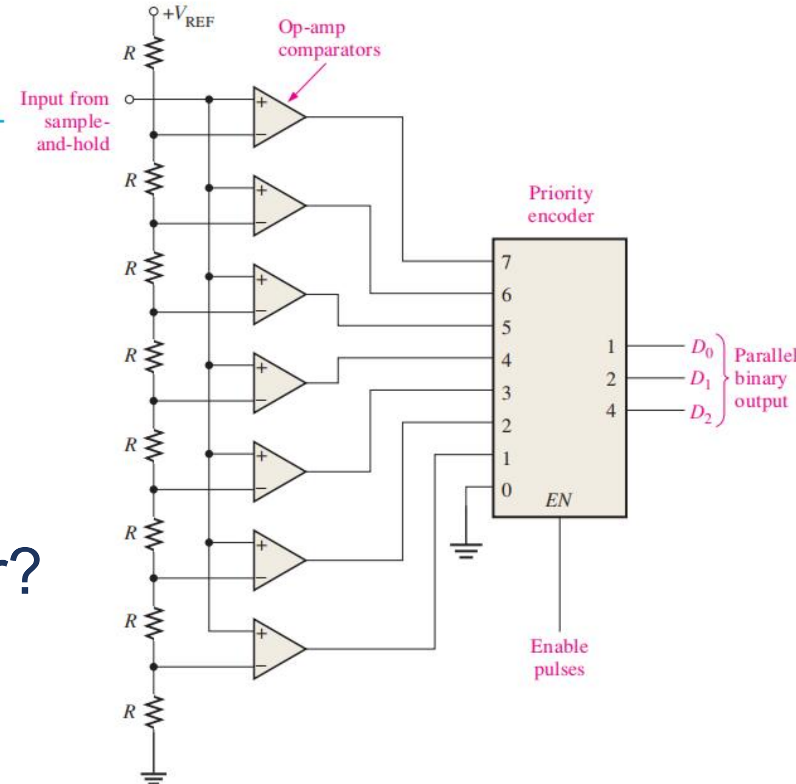
=constant*(decimal value of binary input)

ADC

ADC 1: Flash ADC

In general, 2^n − 1 comparators are required for conversion to an n-bit binary code





- Needs lots of comparators.
- Ultra-fast
- How to design a priority encoder?

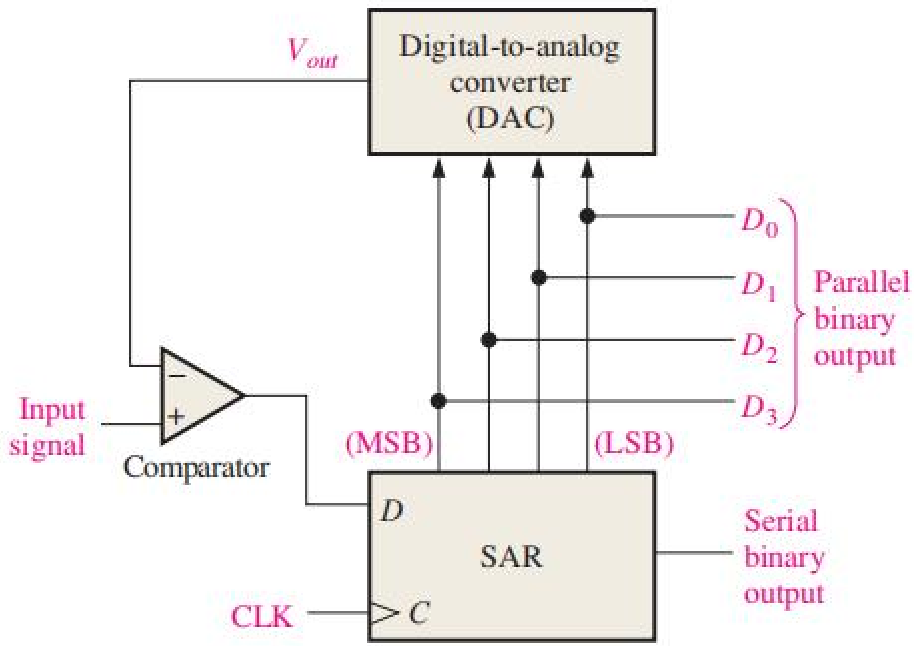
Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

V: valid or not

SAR: Successive Approximation Register.

ADC 1:
Successive-Approximation ADC

The input bits of the DAC are enabled (made equal to a 1) one at a time. So that Vout is getting close to Vin.



SAR: Successive Approximation Register.

ADC 2: Successive-Approximation ADC

Assume Vref=15V

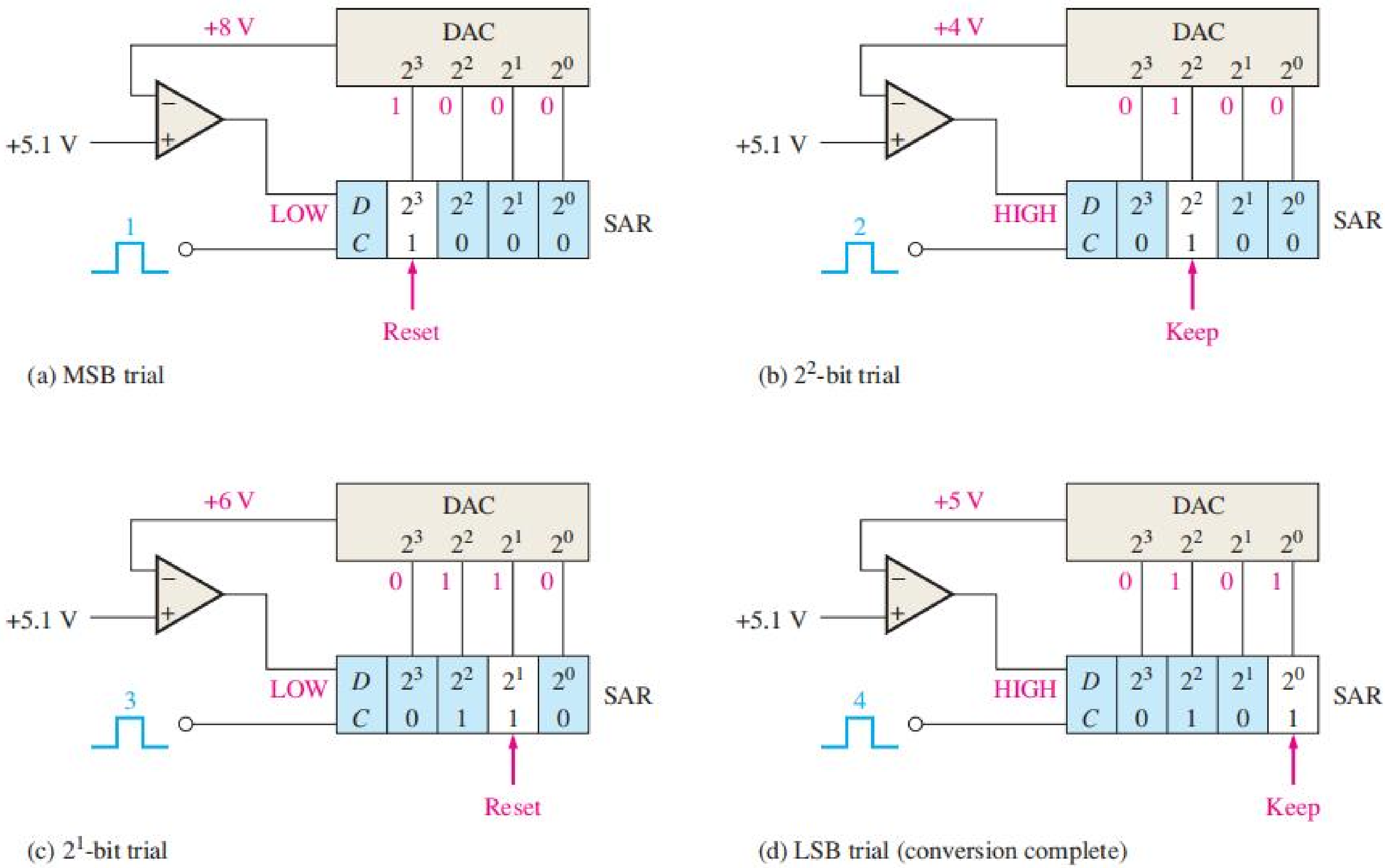


FIGURE 12–18 Illustration of the successive-approximation conversion process.

上海科技大学 ShanghaiTech University

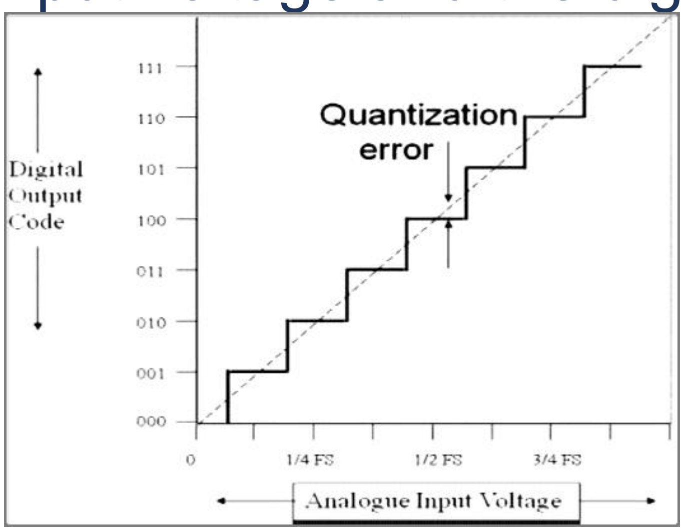
This is like binary search

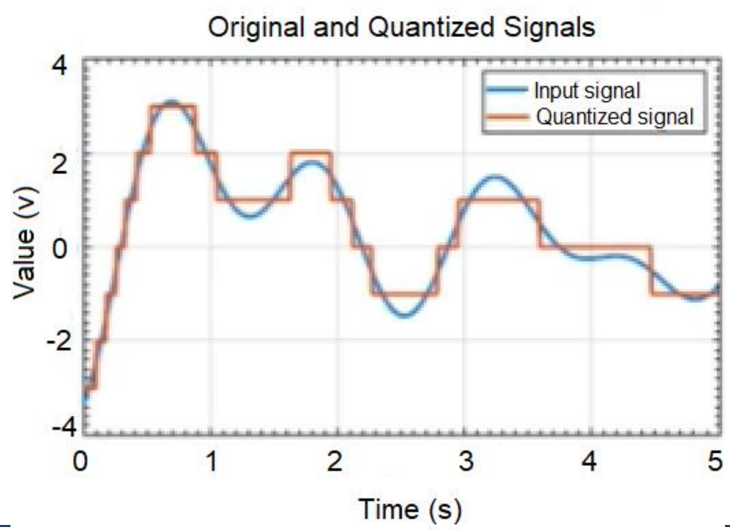
Quantization

- Quantization is the process of mapping a continuous range of analog input values to a finite set of digital levels.
- Resolution: with N bits, the ADC/ADC can represent 2^n distinct levels.

$$\text{LSB (or resolution)} = \frac{V_{ref}}{2^N}$$

- Quantization error: difference between the actual analog input voltage and the digitized output voltage produced.





上海科技大学
ShanghaiTech University

Quantization

Suppose you have a 3-bit ADC with:

- $V_{ref} = 8\,\mathrm{V}$
- Number of levels = $2^3 = 8$
- LSB = $\frac{8\,\mathrm{V}}{8} = 1\,\mathrm{V}$

Then:

- Input voltage of 2.6 V gets quantized to 3 V (nearest digital level).
- Quantization error = 3 V – 2.6 V = **+0.4 V**
- The maximum possible error = ±0.5 V (which is ±½ LSB)

Fourier Transform

Fourier transform is a tool that allows us to represent a real-valued function as a superposition (sum/integral) of sines and cosines with suitably-chosen amplitudes and frequencies

Fourier series

$$g(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)]$$

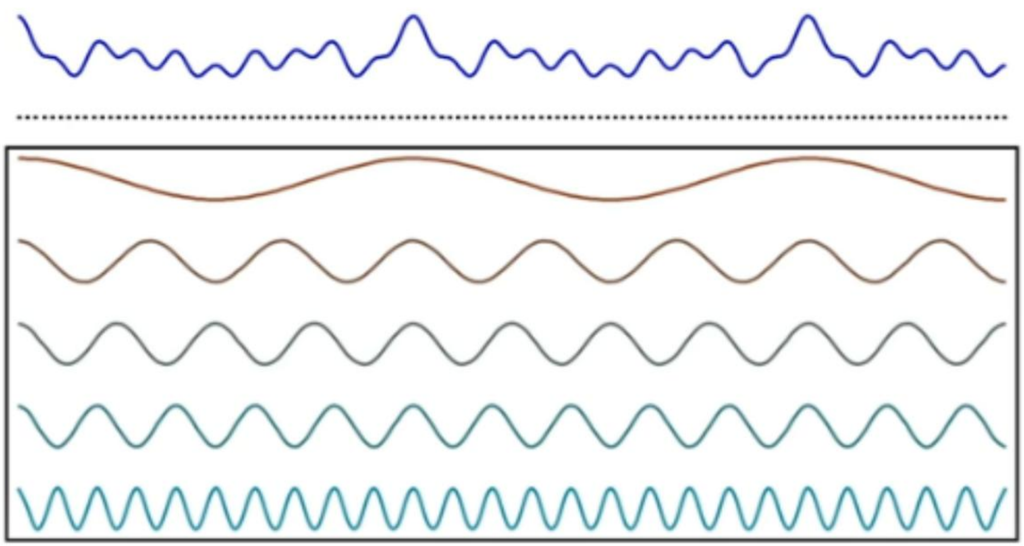
Fourier Coefficients

$$a_n = \frac{2}{T}\int_0^{\boxed{T}} g(t)\cos(nwt)dt$$

$$b_n = \frac{2}{T}\int_0^{T} g(t)\sin(nwt)dt$$

Fourier Transform

$$G\boxed{(f)} = \int_{-\infty}^{\boxed{\infty}} g(t)(\cos(2\pi f t) + i\sin(2\pi f t))dt$$

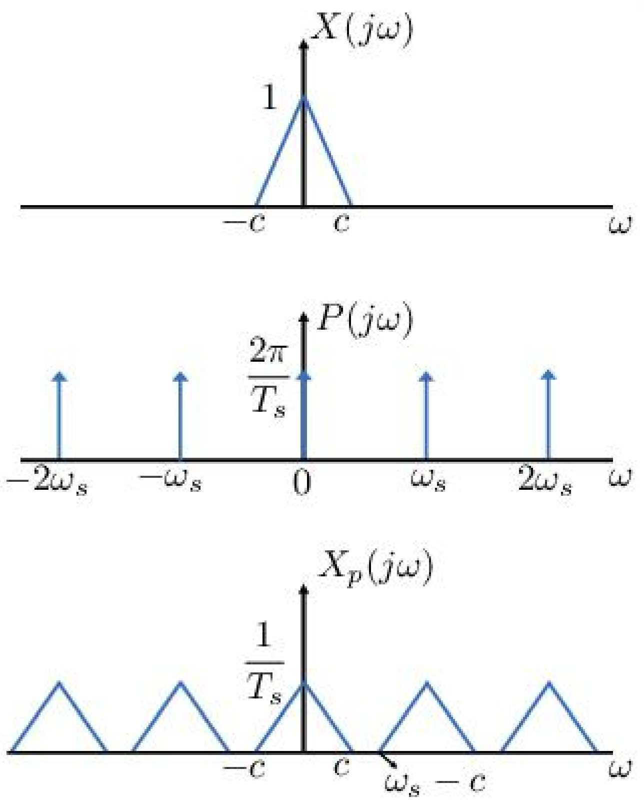


FT: continuous, aperoidic

(If you want to know its principle, take EE150: Signals and Systems)

Nyquist-Shannon Sampling Theorem

An analog signal must be sampled at least twice as fast as the bandwidth of the signal to accurately reconstruct the waveform; otherwise, the high-frequency content creates an alias at a frequency inside the spectrum of interest (passband).



Example:

- Suppose you're sampling audio with frequency content up to 20 kHz.
- To preserve the signal, you must sample at at least 40 kHz (usually 44.1 kHz or 48 kHz in practice).
- An anti-aliasing low-pass filter with a cutoff around 20 kHz is added.

(If you want to know its principle, take EE150: Signals and Systems)

What’s next?

Courses:

- Provided by SIST: 数字集成电路、CA、FPGA、嵌入式系统、板级电路设计实践、...
- 一生一芯：https://ysyx.oscc.cc/

Activities:

- Electronic Competition
- RoboMaster
- InterStudio
- Research Intern in SIST Labs
- ...

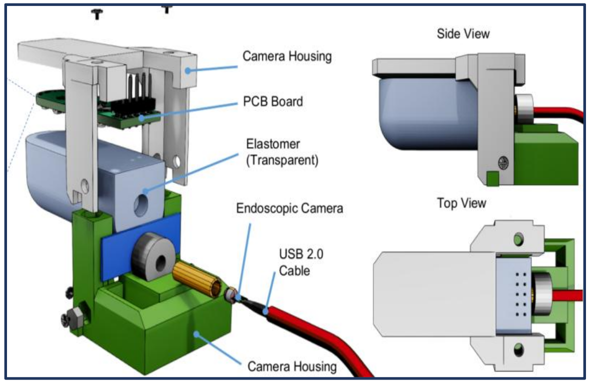
Robot Interaction and Manipulation Laboratory

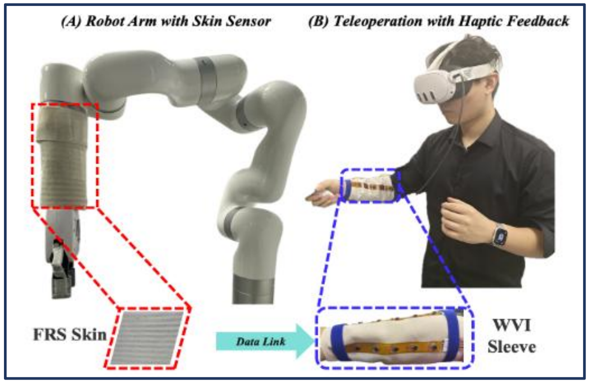
上海科技大学 ShanghaiTech University

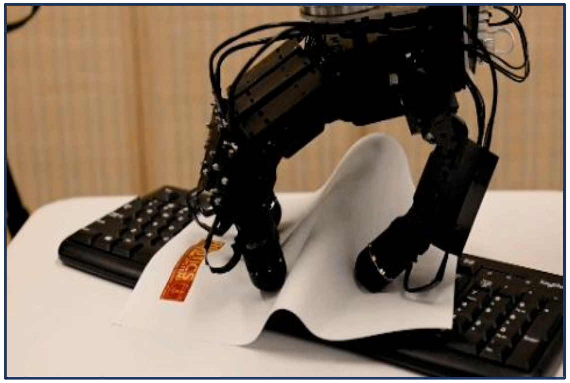
Our lab has openings for undergraduate internship students

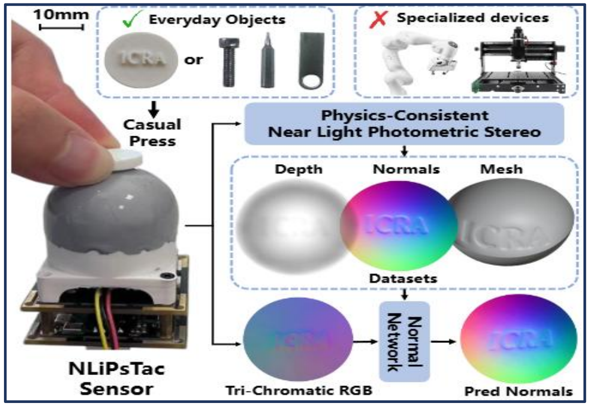
Research Interests: Tactile Perception, Robot Interaction, Robot Manipulation

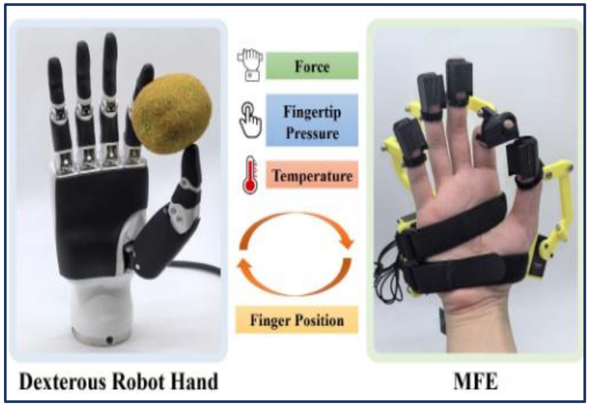
Location: SIST 1D-301A 1D-303, inquiry via email: xiaochx@shanghaitech.edu.cn

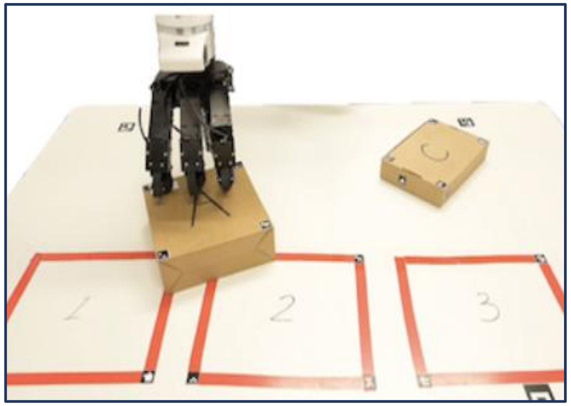












Tactile Perception **Robot Interaction** **Manipulation**